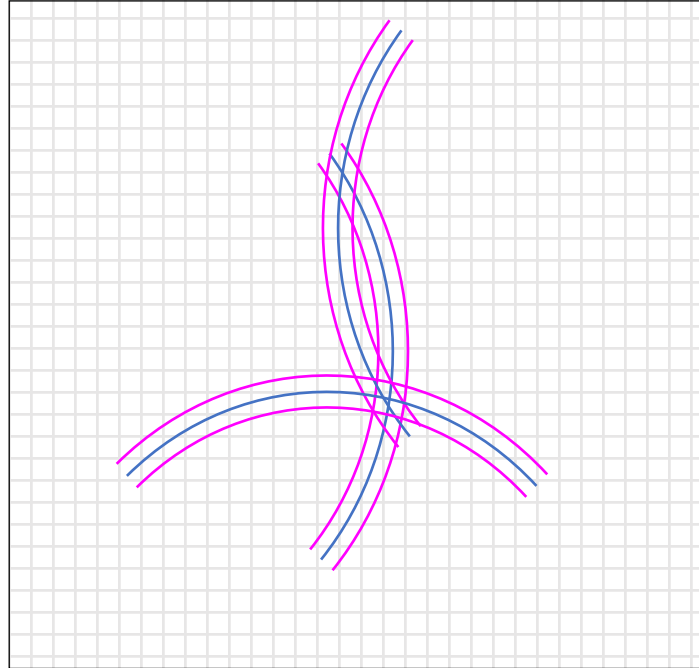
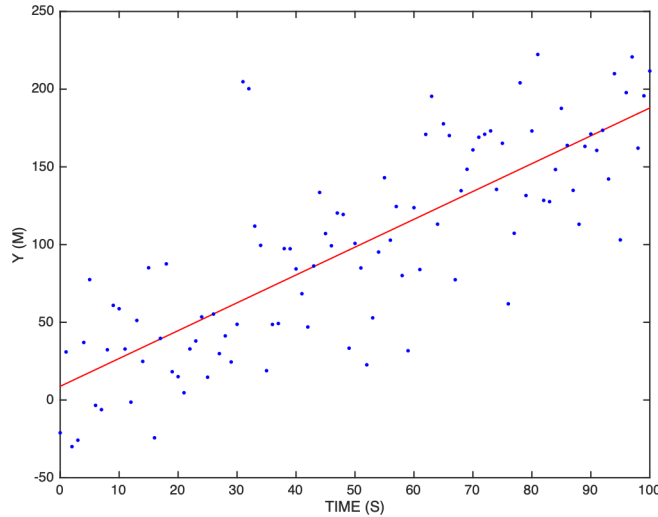


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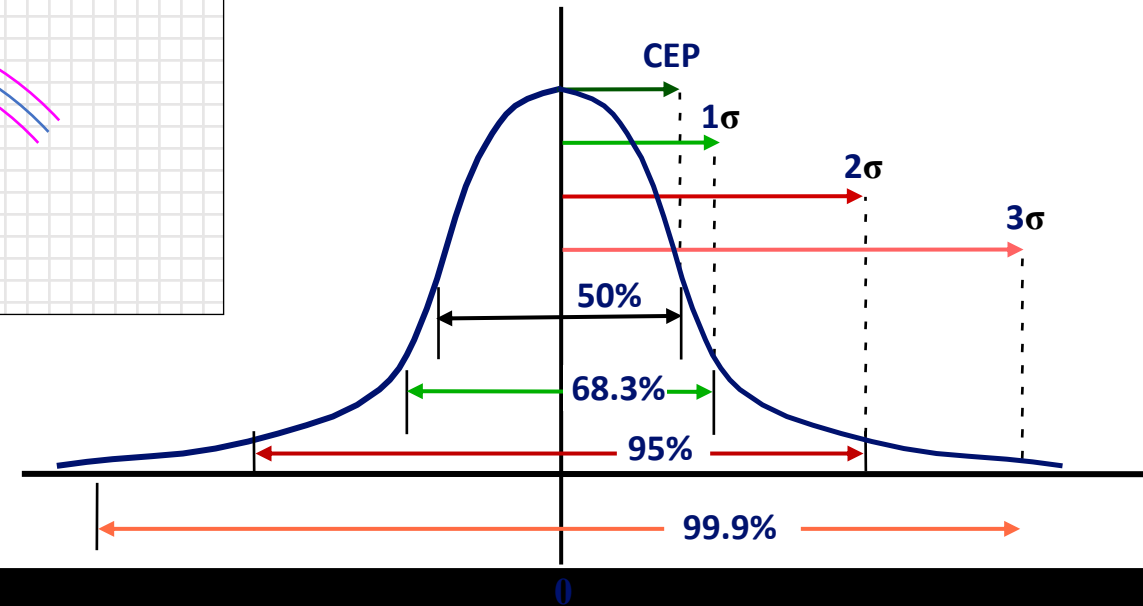
Least Squares Solutions & Error Characterization



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Least Squares

- Least squares will produce a mathematical solution to the graphical solution we associate with GPS – i.e. the intersection of three spheres (where the receiver clock bias has also been solved for and corrected)
- It also provides an approximate solution to an overdetermined set of pseudorange measurement equations that minimizes the sum of the squares of the residuals.
- First we'll review least squares concepts for linear equations then move to the nonlinear GPS case.

Outline

- Linear Model
- Least Squares solution
- Residuals
- Uncertainty/Error Covariance
- Weighted Least Squares solution
- Characterizing positioning errors – 1D, 2D, 3D

Textbook Ch 6 including 6.A



Least Squares Assumptions

- A mathematical model that describes the observations (data) as a linear function of one or more parameters.
- Observation errors that are zero mean and normally distributed.
- The goal is to find the best estimate of the unknown parameters given the observations and knowledge of their error characteristics.



Gaussian or Normal Distribution

$$pdf(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

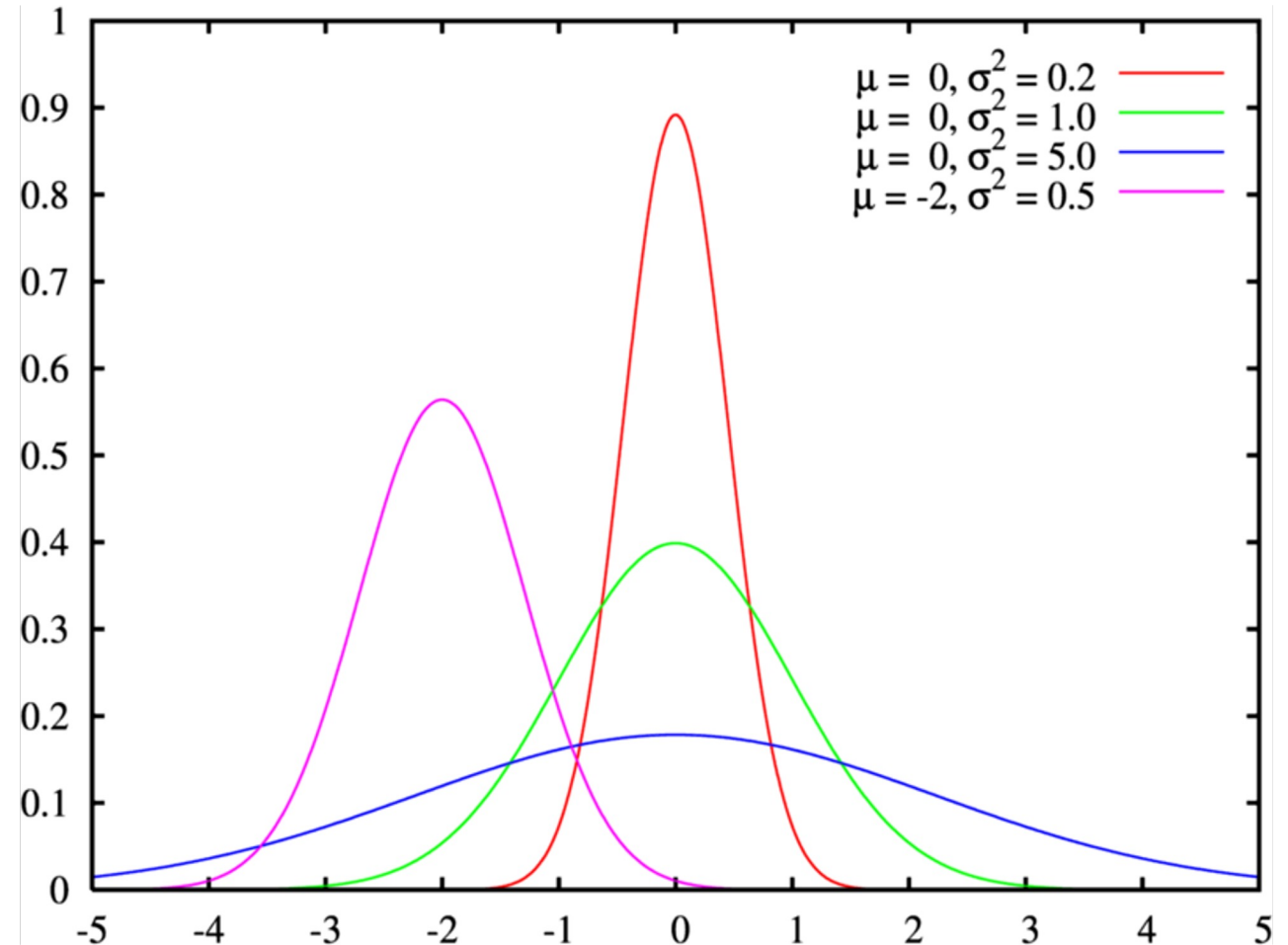
where μ is the mean value
and σ^2 is the variance

For a normal distribution:

68% of the points are within 1σ

95% of the points are within 2σ

99.7% of the points are within 3σ



Expected Value

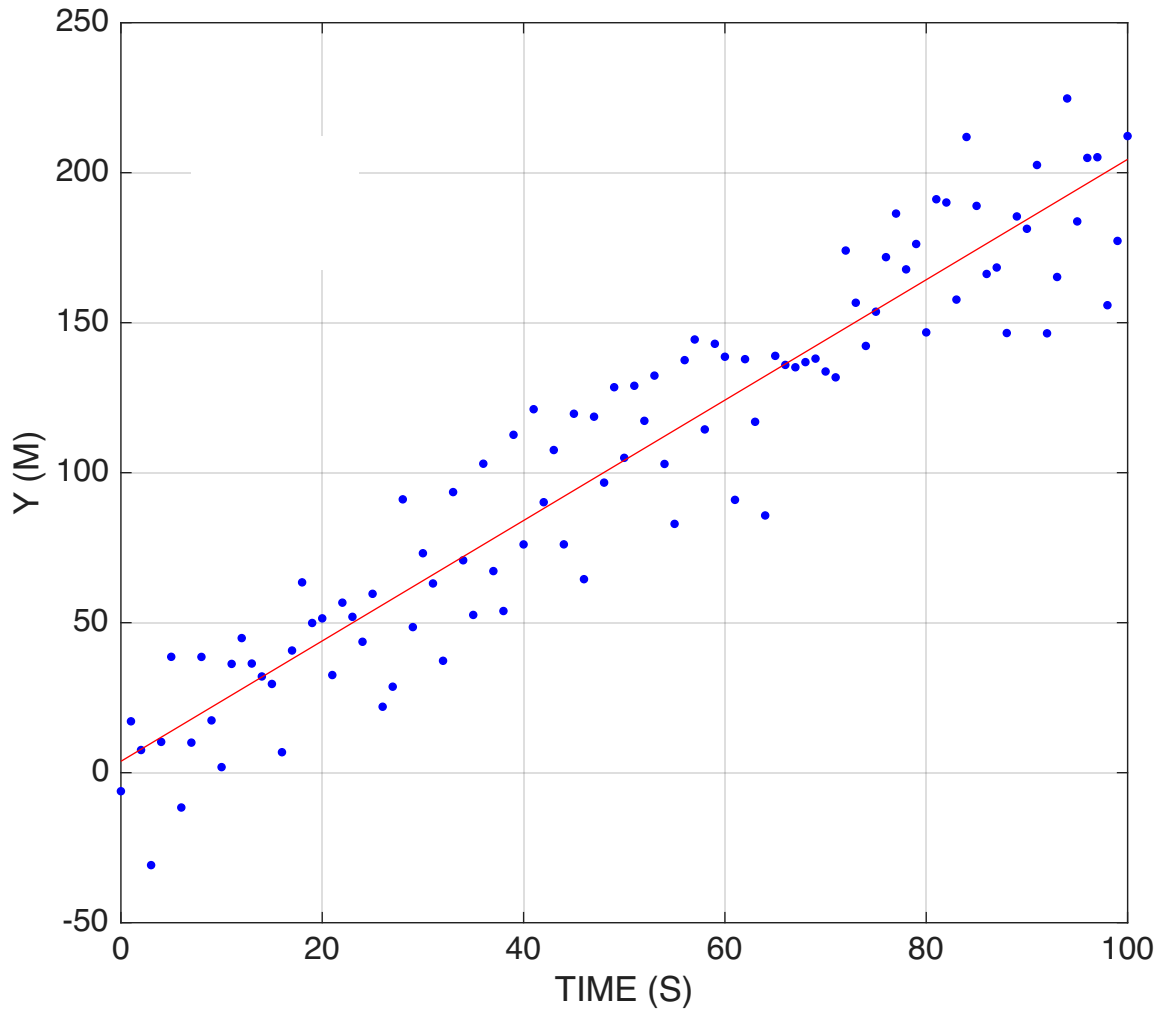
$$E[g(X)] = \int_{-\infty}^{\infty} g(x)f(x)dx$$

$$\bar{x} \triangleq E[X] = \int_{-\infty}^{\infty} xf(x)dx$$

$$\sigma_X^2 \triangleq E[(X - \bar{x})^2] = \int_{-\infty}^{\infty} (x - \bar{x})^2 f(x)dx$$

Where $f(x)$ is the probability density function

Linear Model



$$y = mt + b + \varepsilon$$

y is the observation,

m is slope,

b is offset,

ε is model misfit

$$x \equiv \begin{bmatrix} m \\ b \end{bmatrix}$$

$$y = Ax + \varepsilon$$

$$\text{where } A = \begin{bmatrix} t & 1 \end{bmatrix}$$

and the misfit or error, ε

is assumed to be zero mean

with variance σ^2

Least Squares Solution

- Accumulate all measurements into a matrix equation

$$y = mt + b + \varepsilon$$

$$x \equiv \begin{bmatrix} m \\ b \end{bmatrix}$$

$$y = Ax + \varepsilon$$

$$E[\varepsilon] = \mathbf{0}$$

$$E[\varepsilon\varepsilon^T] = \mathbf{R}$$

$$\text{where } A = \begin{bmatrix} t_1 & 1 \\ t_2 & 1 \\ t_3 & 1 \\ \vdots & \vdots \\ t_N & 1 \end{bmatrix}, \quad y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \\ \vdots \\ y_N \end{bmatrix}$$

Least Squares (continued)

$$\varepsilon = y - Ax \quad \text{How do you penalize the misfit?}$$

Define a cost function

$$J(x) = \varepsilon^T \varepsilon = (y - Ax)^T (y - Ax)$$

To find x^* that gives minimum J , set derivative of J w.r.t. x to 0.

$$0 = -2A^T (y - Ax^*)$$

$$A^T y = A^T A x^*$$

so the (unweighted) least squares solution is:

$$x^* = (A^T A)^{-1} A^T y$$

$$\hat{x} = (A^T A)^{-1} A^T y$$

What happens if A is a square matrix?

More common notation is to use $\hat{}$ for estimate, not $*$, even though this can be confused with unit vector notation....



How do you implement this in MATLAB

- You can set up the A matrix, and actually type:

```
xstar = inv(transpose(A)*A)*transpose(A)*y
```

- Or use the MLDIVIDE (backslash)

```
xstar = A\y
```

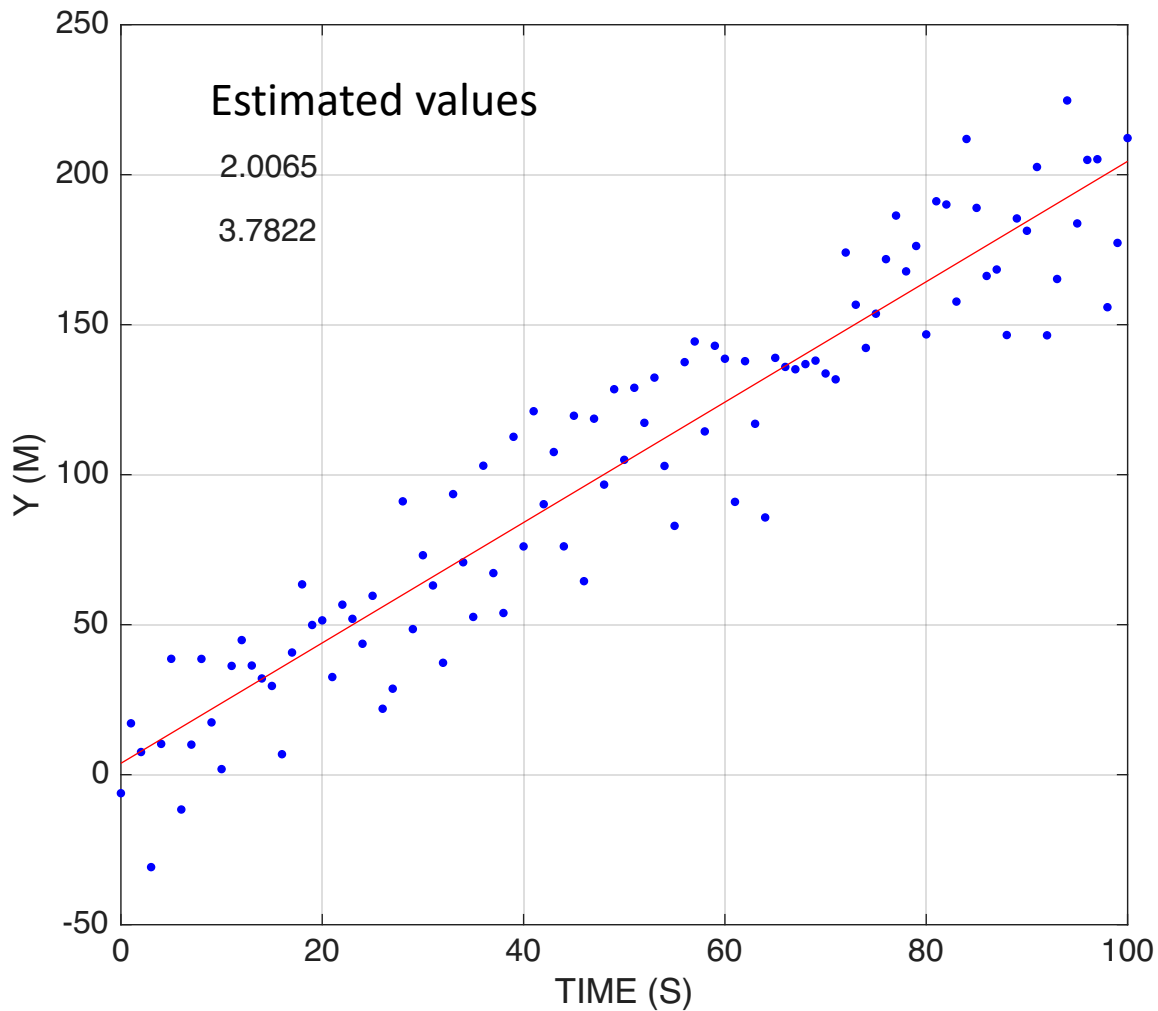
```
Simulate a time series x(t)
t = [0:100]'; st = size(t);
x = 2*t + 5 + randn(st)*20;
```

```
Least squares solution for coefficients
A = [t, ones(st)];
c0 = A\x;
```

```
Evaluate the result
xc = polyval(c0,t);
```

```
plot(t,x,'b.','MarkerSize',8)
    grid on, hold on
plot(t,xc,'r')
xlabel('TIME (S)'), ylabel('Y (M)')
```

Linear Model



$$y = mt + b + \varepsilon$$

y is the observation,

m is slope,

b is offset,

ε is model misfit

$$x \equiv \begin{bmatrix} m \\ b \end{bmatrix}$$

$$y = Ax + \varepsilon$$

$$\text{where } A = \begin{bmatrix} t & 1 \end{bmatrix}$$

and the misfit or error, ε

is assumed to be zero mean

with variance σ^2

Error and uncertainty in the estimates

$$\Delta x \equiv \hat{x} - x = (A^T A)^{-1} A^T y - x = (A^T A)^{-1} A^T \epsilon$$

$$E [\Delta x] = (A^T A)^{-1} A^T E [\epsilon] = \mathbf{0}$$

$$P \equiv E [\Delta x \Delta x^T] = (A^T A)^{-1} A^T R A (A^T A)^{-1}$$

$$\text{If } R = \sigma^2 I, \quad P = \sigma^2 (A^T A)^{-1}$$

G



Data and Estimate Uncertainty

$$\sigma_m = \sigma \sqrt{G(1,1)}$$

If you know measurement noise
sigma, can compute the
theoretical value for estimate

$$y = mt + b + \varepsilon$$

y is the observation, m is slope, b is offset, ε is model misfit

$$x \equiv \begin{bmatrix} m \\ b \end{bmatrix}$$

$$y = Ax + \varepsilon$$

$$\text{where } A = \begin{bmatrix} t & 1 \end{bmatrix}$$

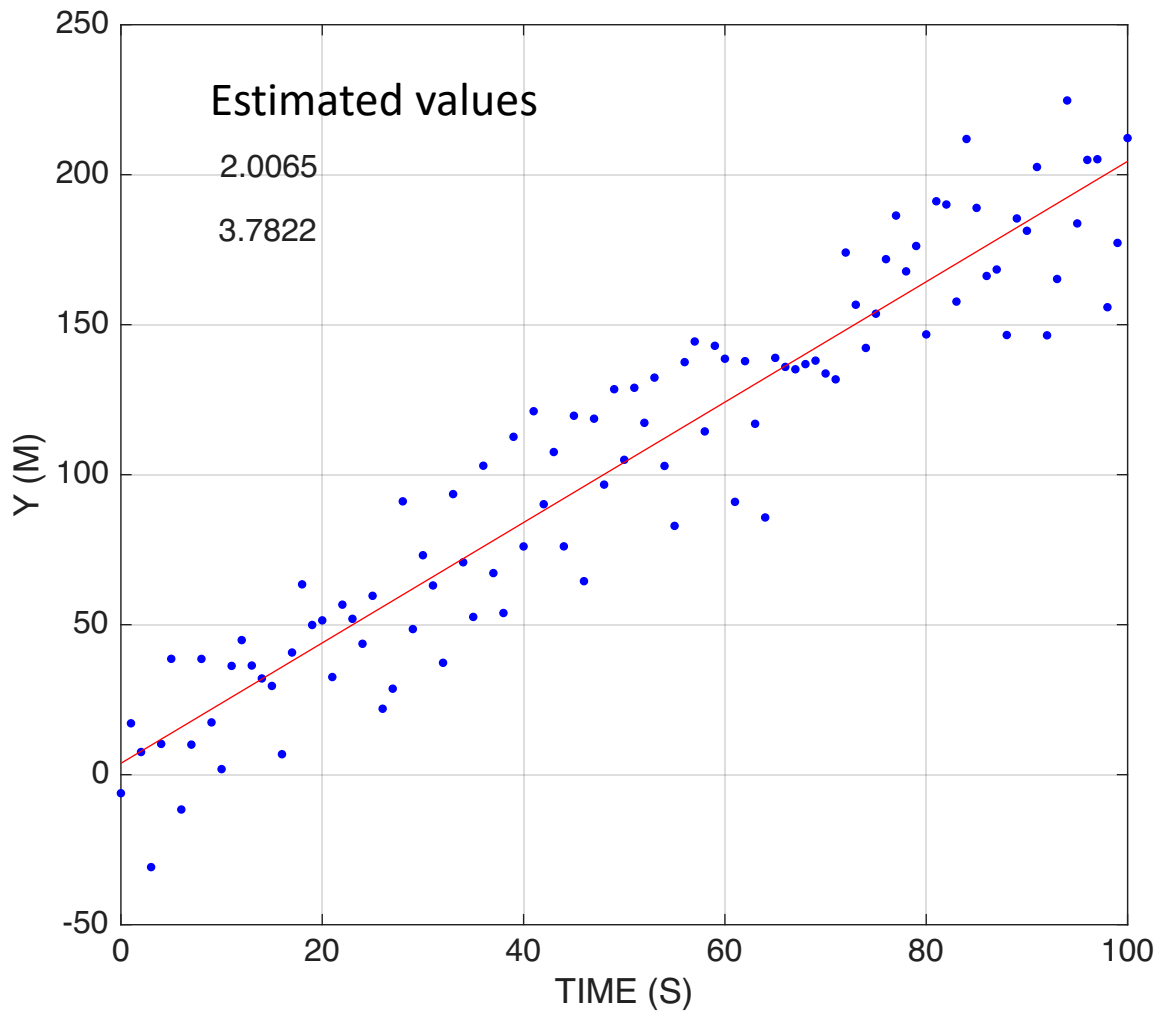
and the misfit or error, e

is assumed to be zero mean

with variance σ^2



Linear Model



P0 =

0.000011648223646 -0.000582411182295
 -0.000582411182295 0.039021549213745

sigc0 =

0.068
 3.9

True values are

2
 5



How do you check your least squares solution?

- Postfit residuals tell how well the LS solution & model fit the data

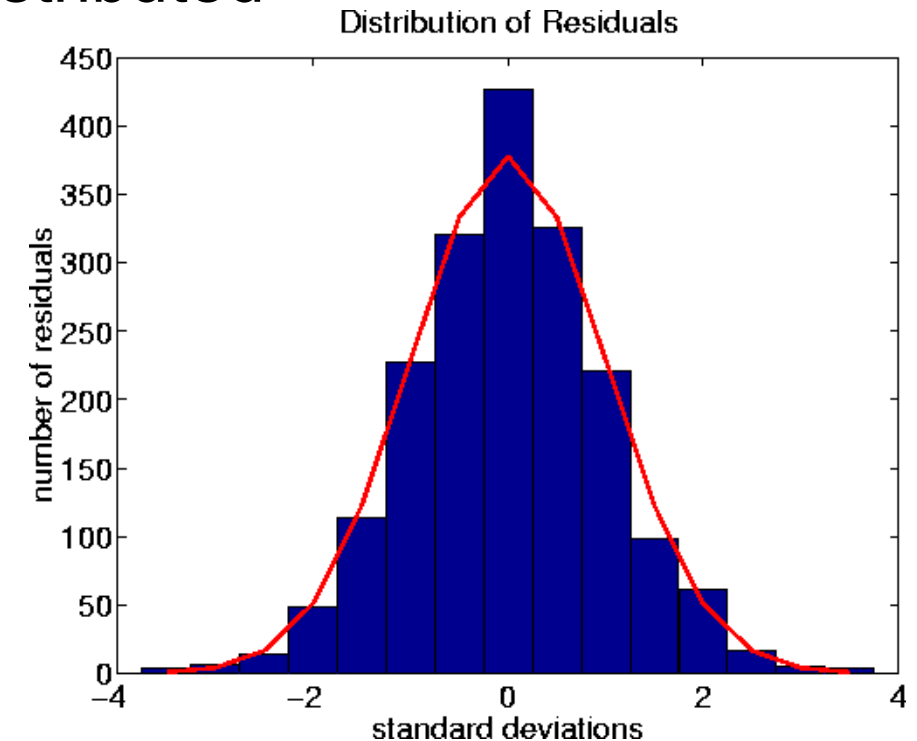
$$pfr = y - A\hat{x}$$

- Residuals should be zero mean and normally distributed

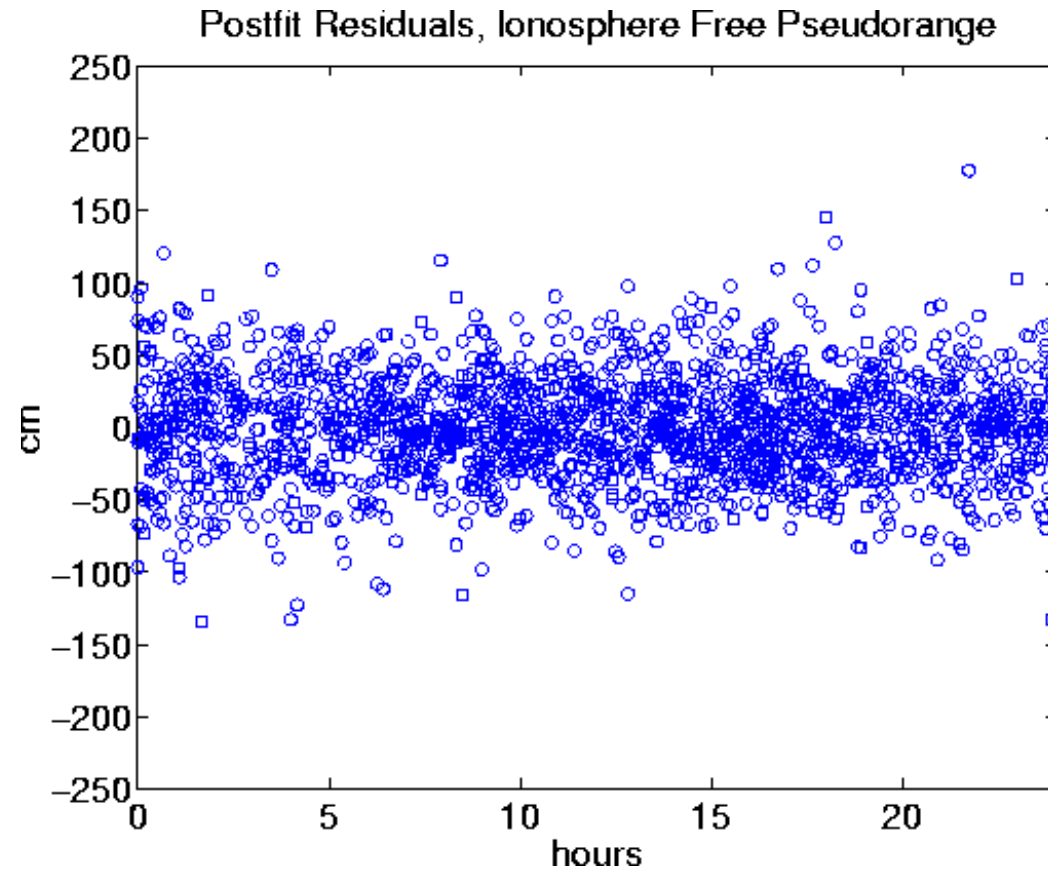
- Number of points in bin x

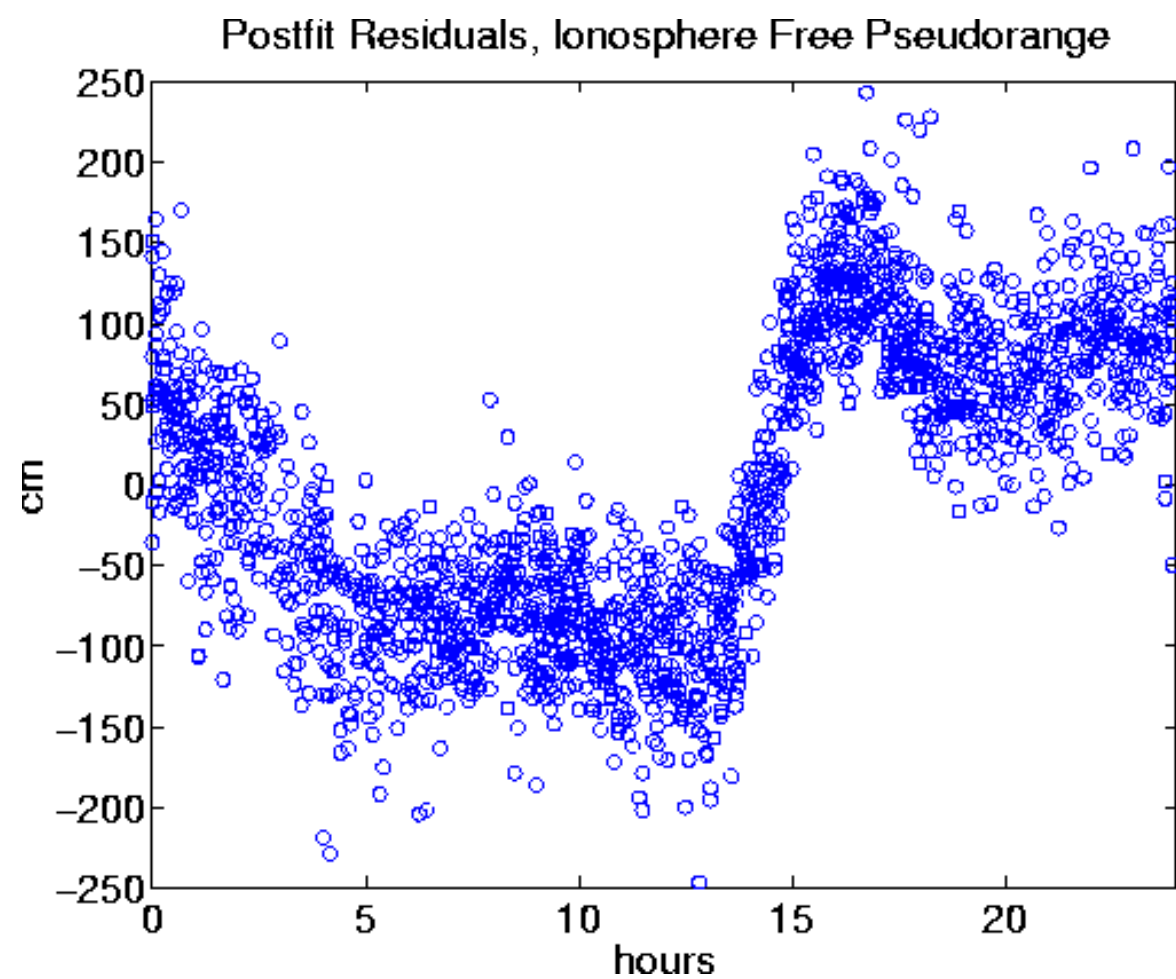
$$N \frac{\Delta x}{\sqrt{2\pi}} e^{-x^2/2}$$

N is total number of points and Δx is bin size

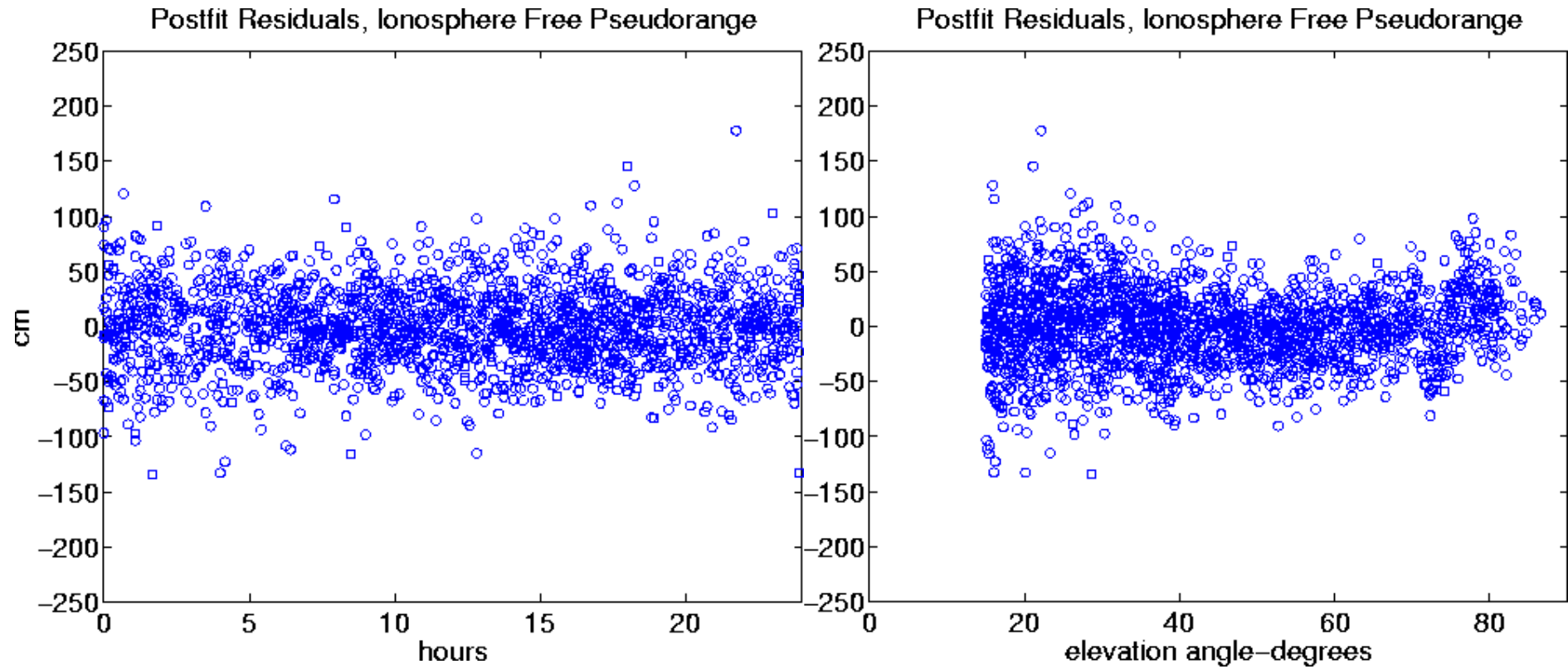


Residual examples

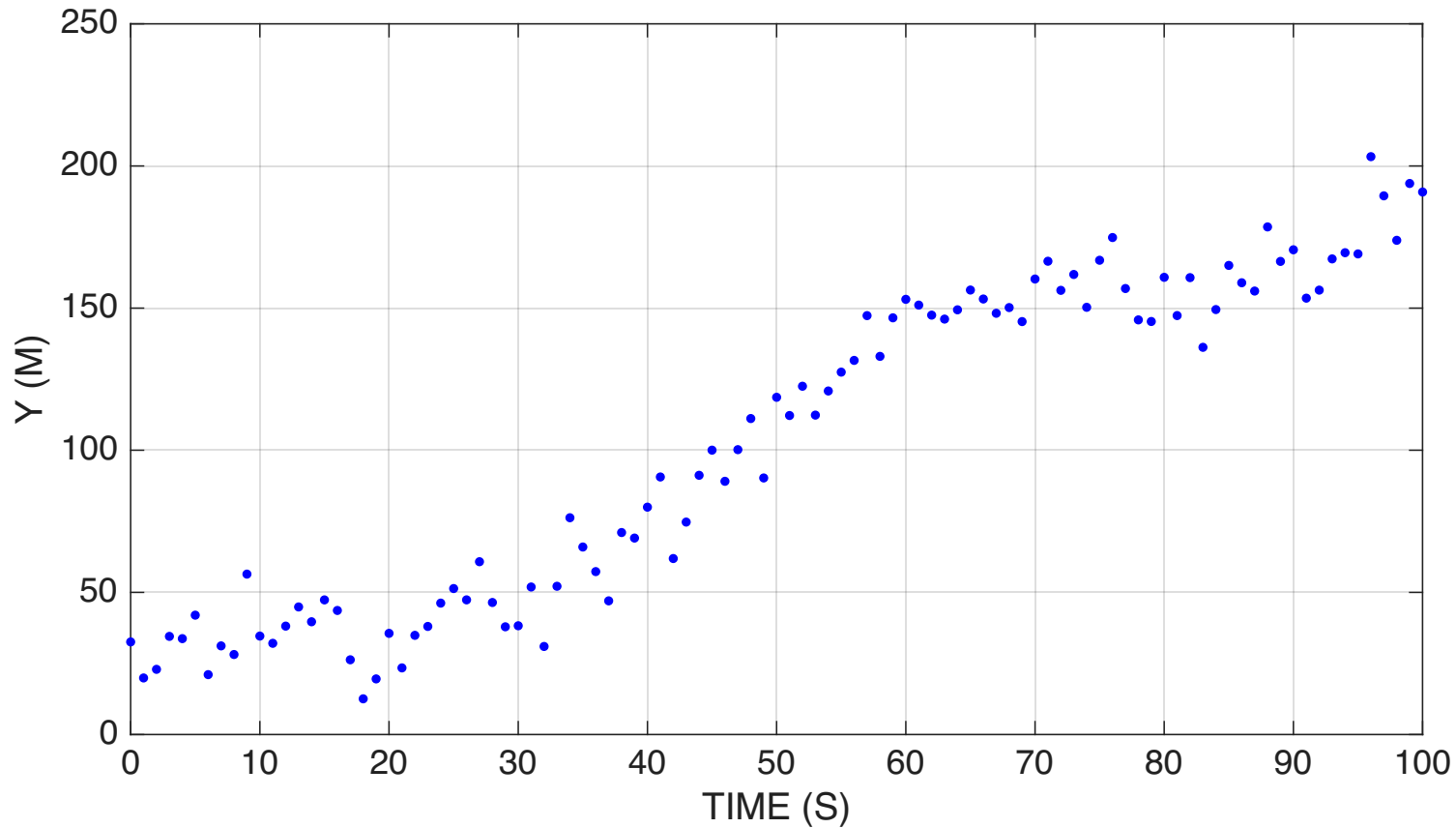




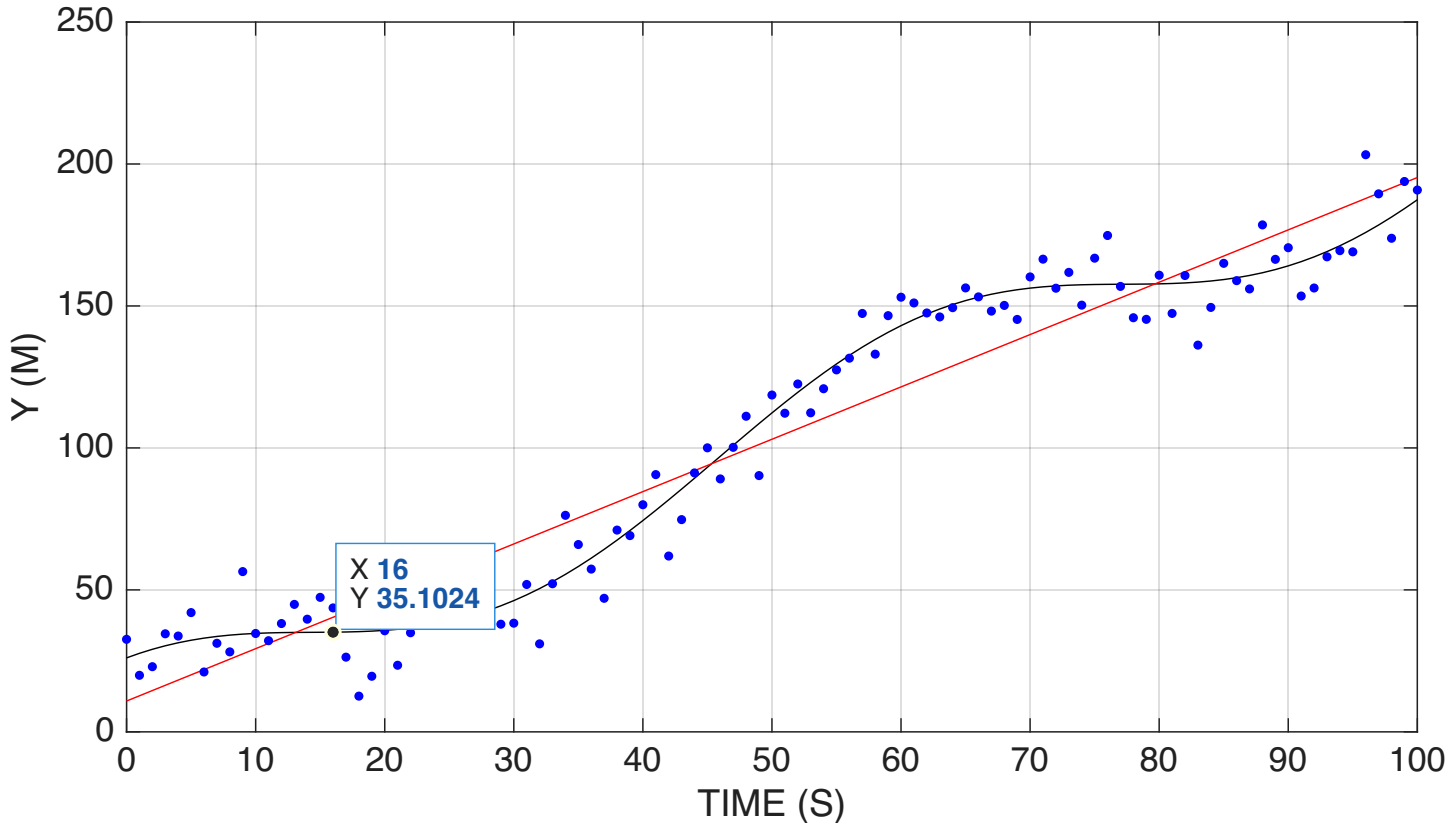
Same Residuals



Can you find a least squares fit to this:



Try different models if you know something about the system



```
f = 0.1;  
y = 5+ 2*t + 20*cos(f*t+0.2)+10*randn(st);
```

```
A = [t, ones(st)];  
c1 = A\y;  
yc1 = A*c1;
```

```
A = [ t, ones(st), cos(f*t), sin(f*t)];  
c2 = A\y;  
yc2 = A*c2;
```



What About Weighted Least Squares?

If each measurement has its own uncertainty, define weight matrix:

$$W = \begin{bmatrix} 1/\sigma_1^2 & 0 & 0 \\ 0 & 1/\sigma_2^2 & 0 \\ 0 & 0 & 1/\sigma_3^2 \end{bmatrix}$$

$$\hat{x} = (A^T W A)^{-1} A^T W y$$

$$P = (A^T W A)^{-1}$$



Transforming estimates and uncertainty between ECEF and ENU

Original estimate

$$\hat{x} = \begin{bmatrix} \Delta X_r \\ \Delta Y_r \\ \Delta Z_r \\ R_{clk} \end{bmatrix}$$

What we would
rather have

$$\hat{x}_{env} = \begin{bmatrix} \Delta E_r \\ \Delta N_r \\ \Delta V_r \\ R_{clk} \end{bmatrix}$$

Transformation matrix

$$\phi_{env} = \begin{bmatrix} -\sin \lambda & \cos \lambda & 0 & 0 \\ -\sin \varphi \cos \lambda & -\sin \varphi \sin \lambda & \cos \varphi & 0 \\ \cos \varphi \cos \lambda & \cos \varphi \sin \lambda & \sin \varphi & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$P_{ENU} = \phi_{ENU} (P_{ECEF}) \phi_{ENU}^T$$



CHARACTERIZING POSITIONING ERRORS

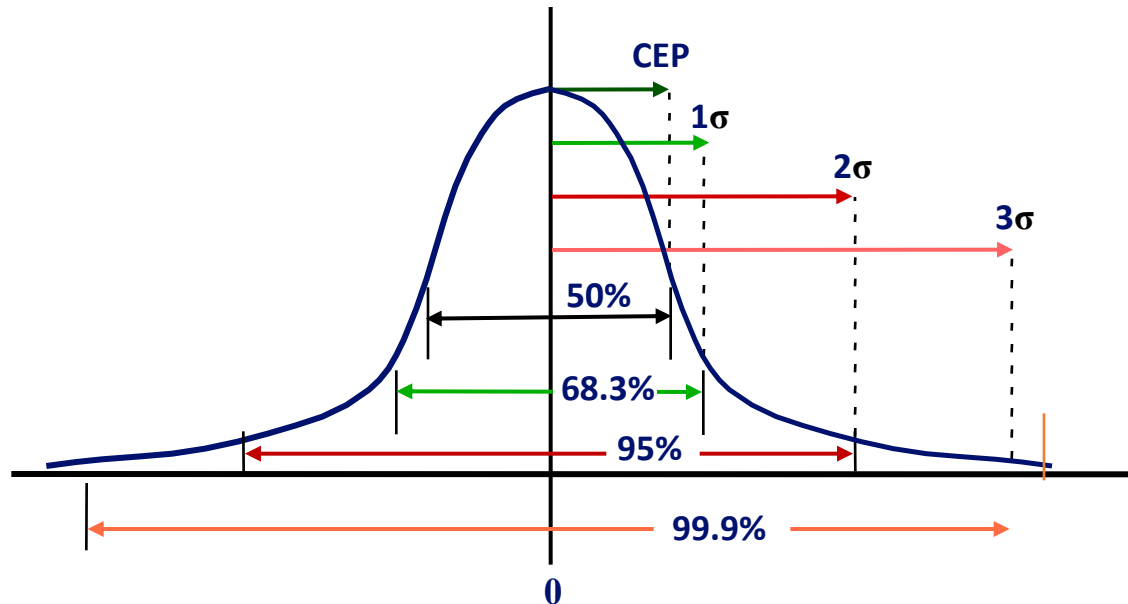
- Error measures used in positioning come from different positioning requirements.
 - In navigation and positioning, two dimensional distributions are of interest for horizontal positioning.
 - Three dimensional errors are also important, although very often the vertical direction has very different performance requirements and is specified separately.
-
- Normal or Gaussian distribution
 - Scalar Errors – mean, standard deviation, RMS
 - 2D Errors – standard deviation, 2DRMS, CEP
 - 3D Errors - standard deviation, SEP

Accuracy Characterization

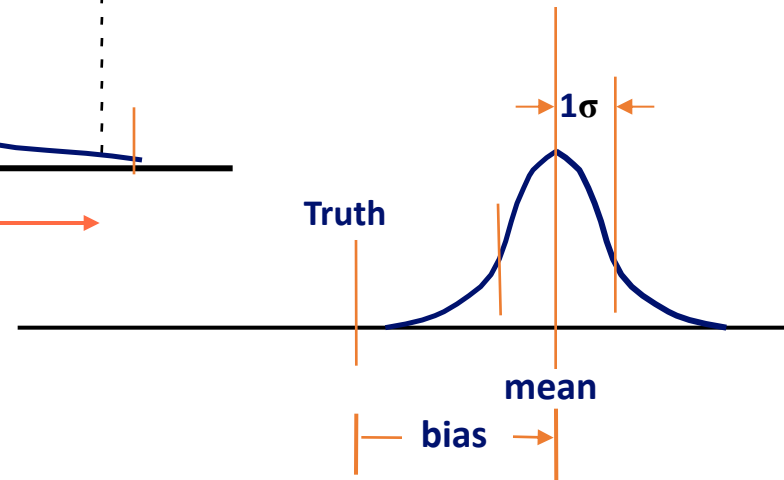
Normal Statistical Distribution

$$pdf(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

μ is the mean value and
 σ^2 is the variance



Zero mean
Normal Distribution



Biased Normal Distribution

Scalar Errors

The mean value over n measurements is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = E[x_i]$$

The deviation of a measurement from the mean is $\delta x_i = x_i - \bar{x}$

The standard deviation is:

For a scalar rv with normal distribution, the probability of being within 1σ of the mean is 68.3%

$$\sigma = \sqrt{\frac{\sum_{i=1}^n \delta x_i^2}{n-1}}$$

The variance is σ^2

The root mean square value is:

$$RMS = \sqrt{\frac{\sum_{i=1}^n x_i^2}{n}}$$



2-D Errors

The mean value over n measurements is:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i = E[x_i] \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i = E[y_i]$$

The deviation of a measurement from the mean is $\delta x_i = x_i - \bar{x}$ $\delta y_i = y_i - \bar{y}$

The covariance is:

$$P = \begin{bmatrix} \sigma_x^2 & \frac{\sum_{i=1}^n \delta x_i \delta y_i}{n-1} \\ \frac{\sum_{i=1}^n \delta x_i \delta y_i}{n-1} & \sigma_y^2 \end{bmatrix}$$

- The covariance matrix, P, defines the error ellipse.
- Eigenvalues are squares of semimajor and semiminor axes of the ellipse
- Eigenvectors show the direction
- For 2D Gaussian with equal sigmas, probability of being inside 1- σ ellipse is ~40%
- For 3D Gaussian with equal sigmas, probability of being inside 1- σ ellipsoid is ~20%

Probability of normally distributed errors being inside ellipsoid

$$(x - \bar{x})^T P^{-1} (x - \bar{x}) = L^2$$

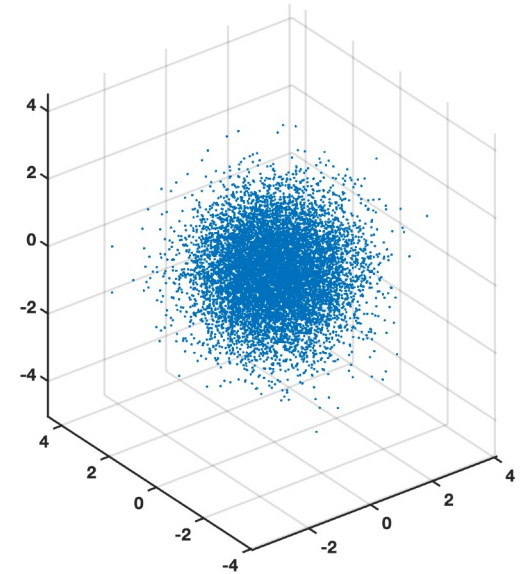
n	$1-\sigma$	$2-\sigma$	$3-\sigma$
1	0.6827	0.9545	0.9973
2	0.3935	0.8647	0.9889
3	0.1987	0.7385	0.9707

For dimensions $n=1, 2, 3$

$$n = 1: \sqrt{2/\pi} \int_0^L \exp(-\frac{1}{2}r^2) dr = \operatorname{erf}\left(\frac{L}{\sqrt{2}}\right)$$

$$n = 2: \int_0^L \exp\left(-\frac{1}{2}r^2\right) r dr = 1 - \exp\left(-\frac{1}{2}L^2\right)$$

$$n = 3: \sqrt{2/\pi} \int_0^L \exp\left(-\frac{1}{2}r^2\right) r^2 dr = \operatorname{erf}\left(\frac{L}{\sqrt{2}}\right) - \sqrt{2/\pi} L \exp\left(-\frac{1}{2}L^2\right)$$



From Bryson and Ho, *Applied Optimal Control*, 1975

2-D Errors

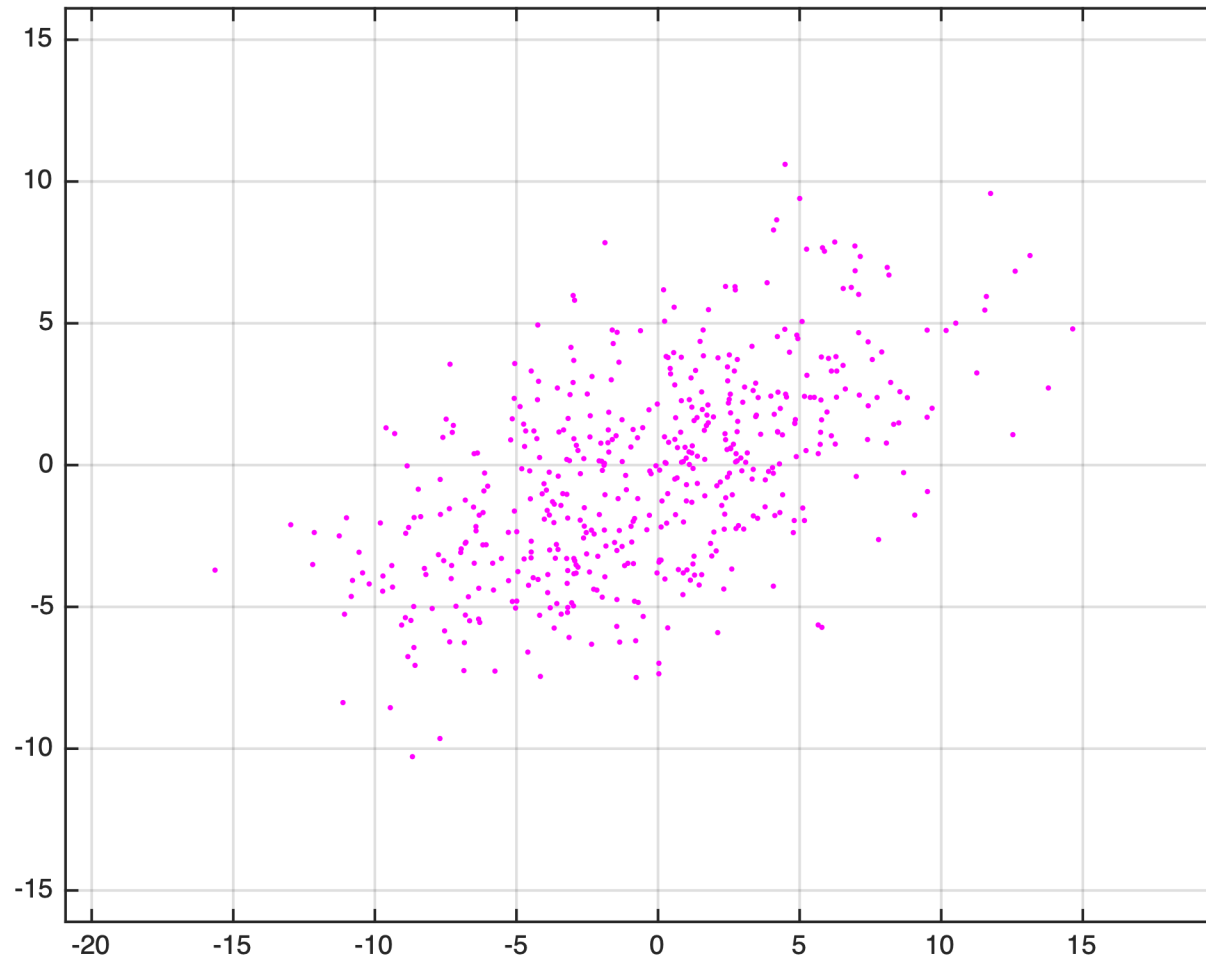
- DRMS - Distance Root Mean Square 2-D $DRMS = \sqrt{\sigma_x^2 + \sigma_y^2}$
 - Also called the **Radial Error, Mean Square Position Error, or RSS**
 - Probability of being inside DRMS circle:
 - *For $\sigma_x = \sigma_y$ probability = 63%*
 - *For $\sigma_x = 10\sigma_y$ probability = 68%*
- 2 DRMS - Two inconsistent definitions
 - 2 x DRMS : Probability between 95.4% and 98%
(US Federal Radionavigation Plan)
 - Same as DRMS (63-68%) (NATO's Standardization Agreement)

CEP – Circular Error Probable

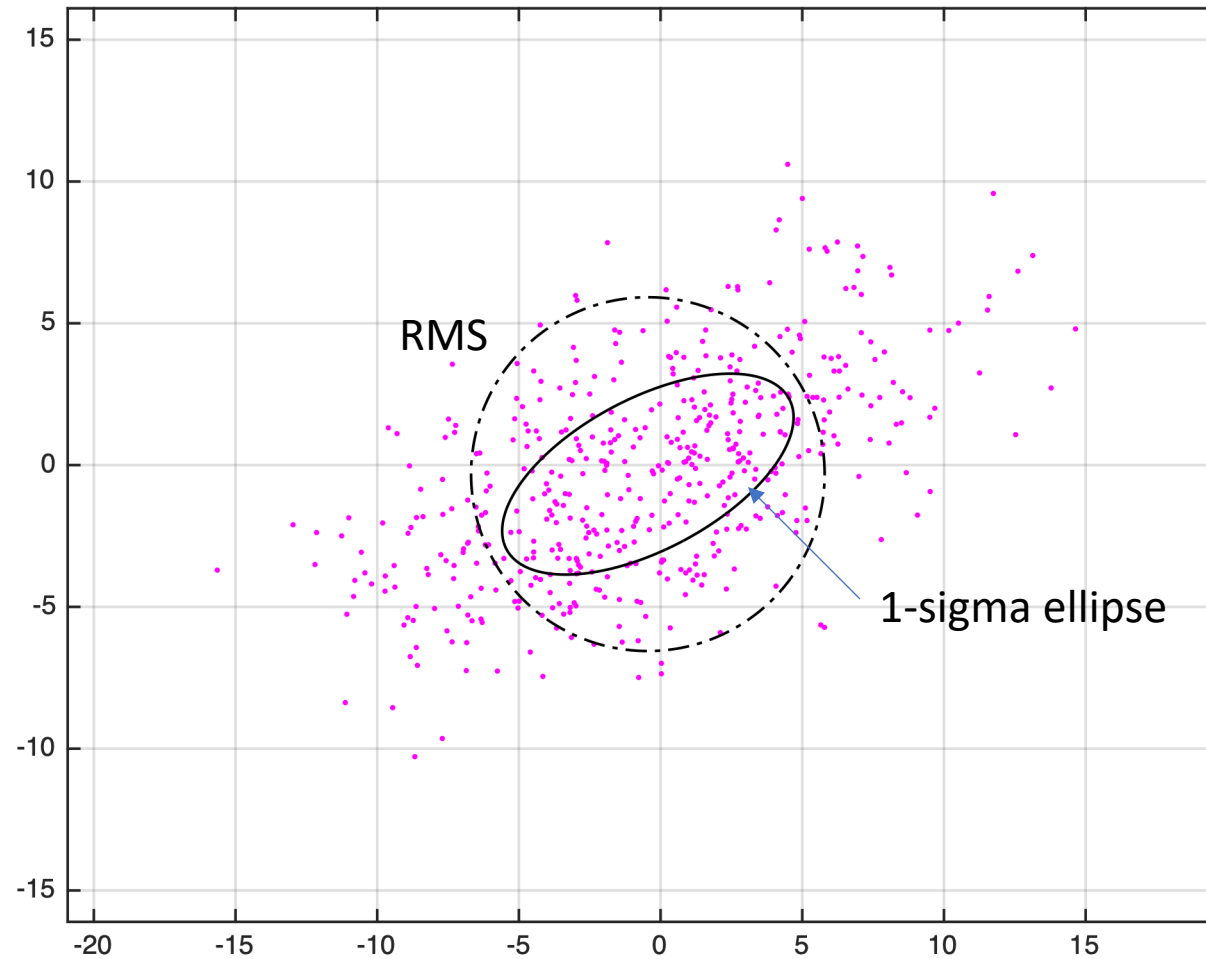
- The radius of a circle inside which 50% of the points fall.
- How does it compare to σ_x and σ_y ?
For a normal distribution
 - For $\sigma_x = \sigma_y$ $CEP = 1.18 \sigma$
 - Approximate : For $\sigma_y / 3 < \sigma_x < 3 \sigma_y$
 $CEP = 0.59 (\sigma_x + \sigma_y) \pm 3\%$
- 95% circle is $CEP \times 2.08 = 2 \times DRMS$
- 99% circle is $CEP \times 2.58$
- Reference: *Aerospace Avionics Systems - A Modern Synthesis*, G. M. Siouris, 1993.



Example of 2D Errors



Example of 2D Errors



Acknowledgements

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Next lecture: GPS Least Squares “*Point Solutions*”

